Mermaid Flowchart Customization Guide

Deep dive into flowchart customization — shapes, colors, layout, styling, and advanced patterns.

Node Shapes

Every shape has semantic meaning. Choose based on the node’s purpose.

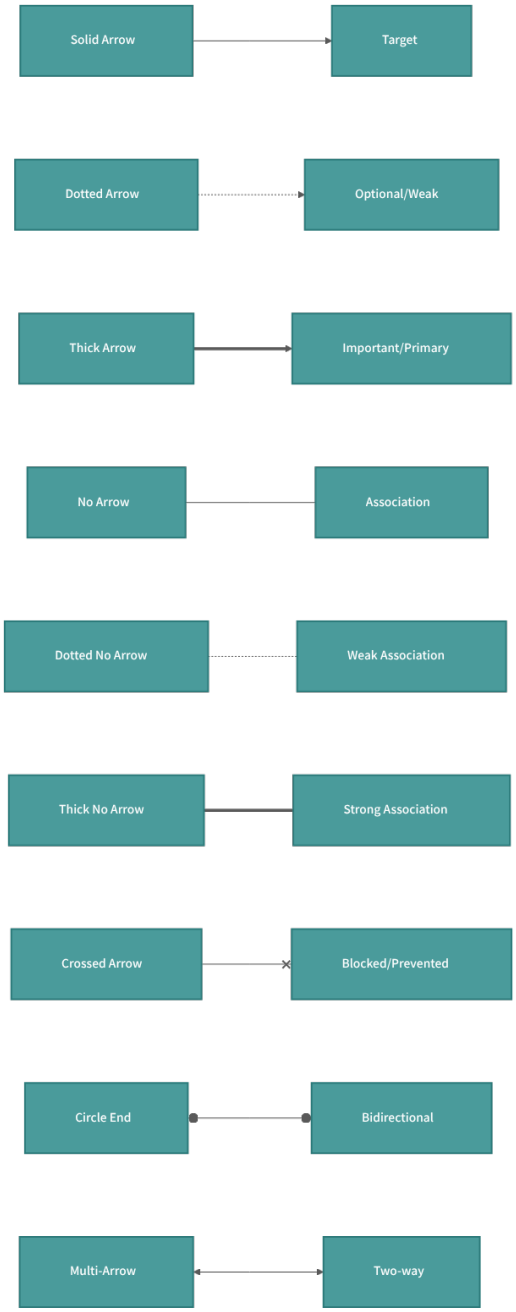
> ***Note:*** *Diagram 1 failed to render (unsupported syntax)*

Shape Syntax Quick Reference:

Shape	Syntax	Use Case
Rectangle	`[text]`	Standard process
Rounded	`([text])`	Start/end points
Stadium	`([text])`	Same as rounded
Subroutine	`[[text]]`	Function call
Cylindrical	`[(text)]`	Database
Circle	`((text))`	Connection point
Double Circle	`(((text)))`	-
Asymmetric	`>text]`	Choice/flag
Rhombus	`{text}`	Decision
Hexagon	`{{text}}`	Preparation
Parallelogram (input)	`[/text/]`	Input data
Parallelogram (output)	`[\text\]`	Output data
Trapezoid	`[/text\]`	Manual operation
Inverted Trapezoid	`[\text/]`	-

Arrow Types and Line Styles

Arrows communicate relationships. Mix types to show different connection meanings.



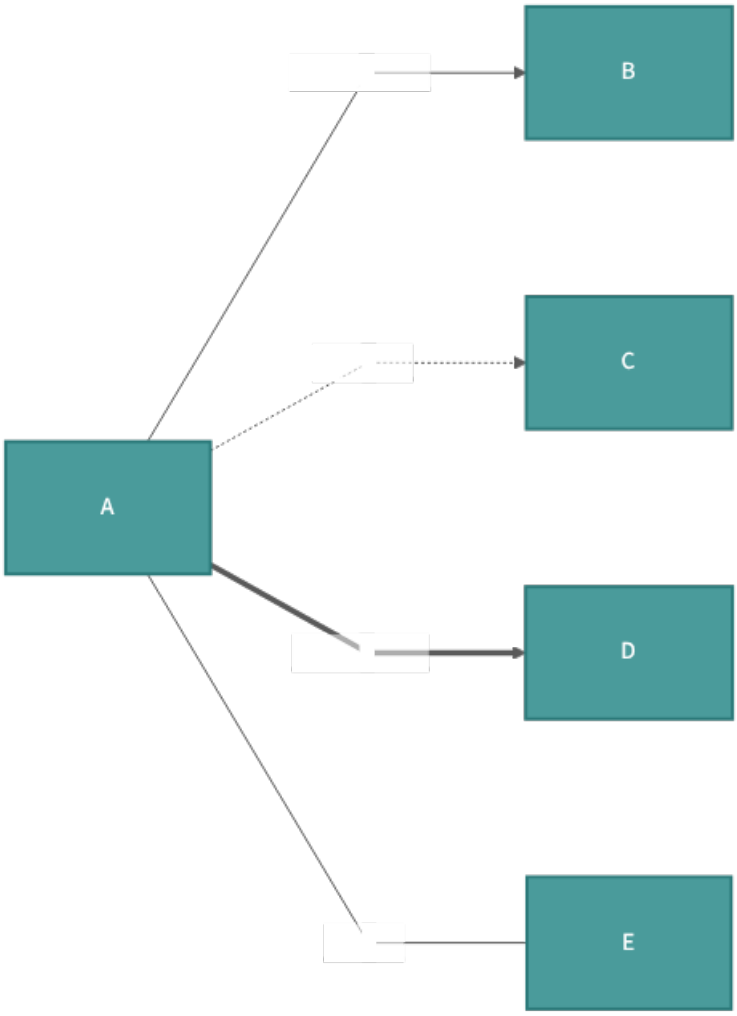
Mermaid Diagram 2

Arrow Syntax:

Pattern	Code	Meaning
Solid line, arrow	`-->`	Standard flow
Dotted line, arrow	`-.->`	Optional/fallback
Thick line, arrow	`==>`	Primary/critical path
Solid line, no arrow	`---`	Association
Dotted line, no arrow	`-.-`	Weak association

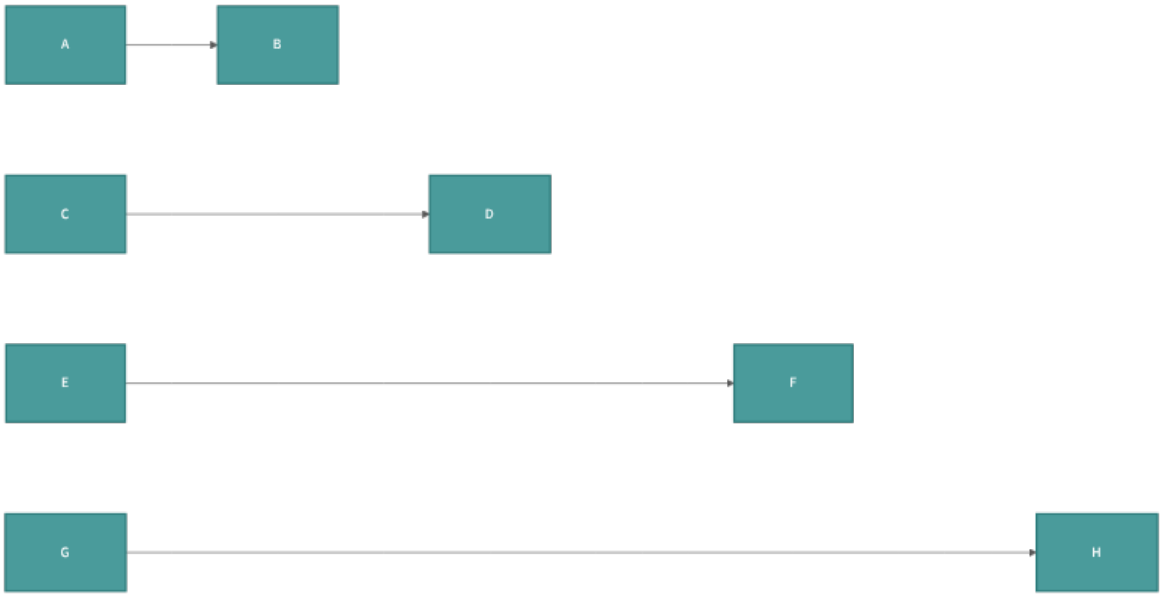
Pattern	Code	Meaning
Crossed	`--x`	Blocked
Circle ends	`o--o`	Bidirectional
Double arrow	`<-->`	Two-way communication

Adding Labels:



Mermaid Diagram 3

Longer Arrows (add dashes):

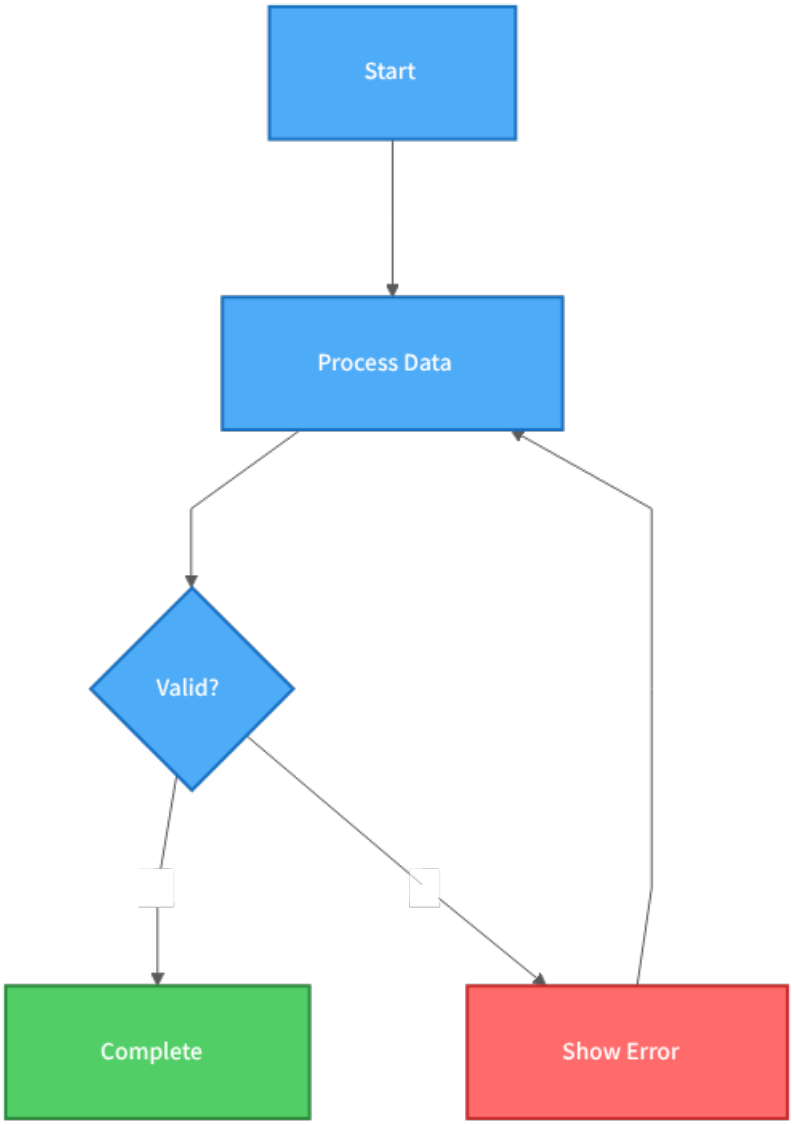


Mermaid Diagram 4

Color and Styling with classDef

`classDef` creates reusable style classes. Apply to multiple nodes for consistency.

Basic classDef Pattern



Mermaid Diagram 5

classDef Syntax:

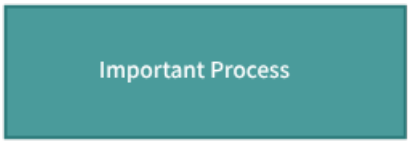
```
classDef <className> <property>:<value>,<property>:<value>,...
class <nodeId> <className>
```

Common Properties:

Property	Values	Example
`fill`	Color hex/rgb	`fill:#ff6b6b`
`stroke`	Border color	`stroke:#c92a2a`
`stroke-width`	Border thickness	`stroke-width:3px`
`color`	Text color	`color:#fff`

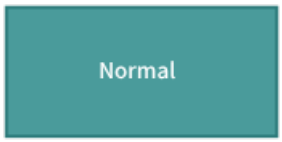
Property	Values	Example
`stroke-dasharray`	Dash pattern	`stroke-dasharray: 5 5`
`rx`	Corner radius	`rx:10px`
`ry`	Corner radius	`ry:10px`

Multiple Classes on One Node



Mermaid Diagram 6

Inline Styling (Quick One-offs)

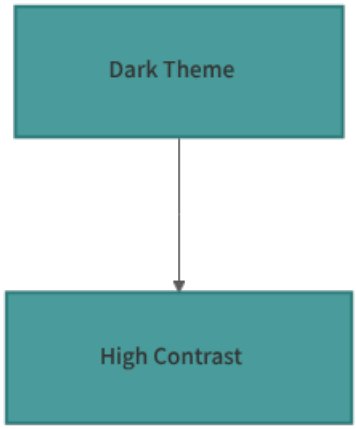


Mermaid Diagram 7

Theme Customization

Apply global themes to entire diagrams.

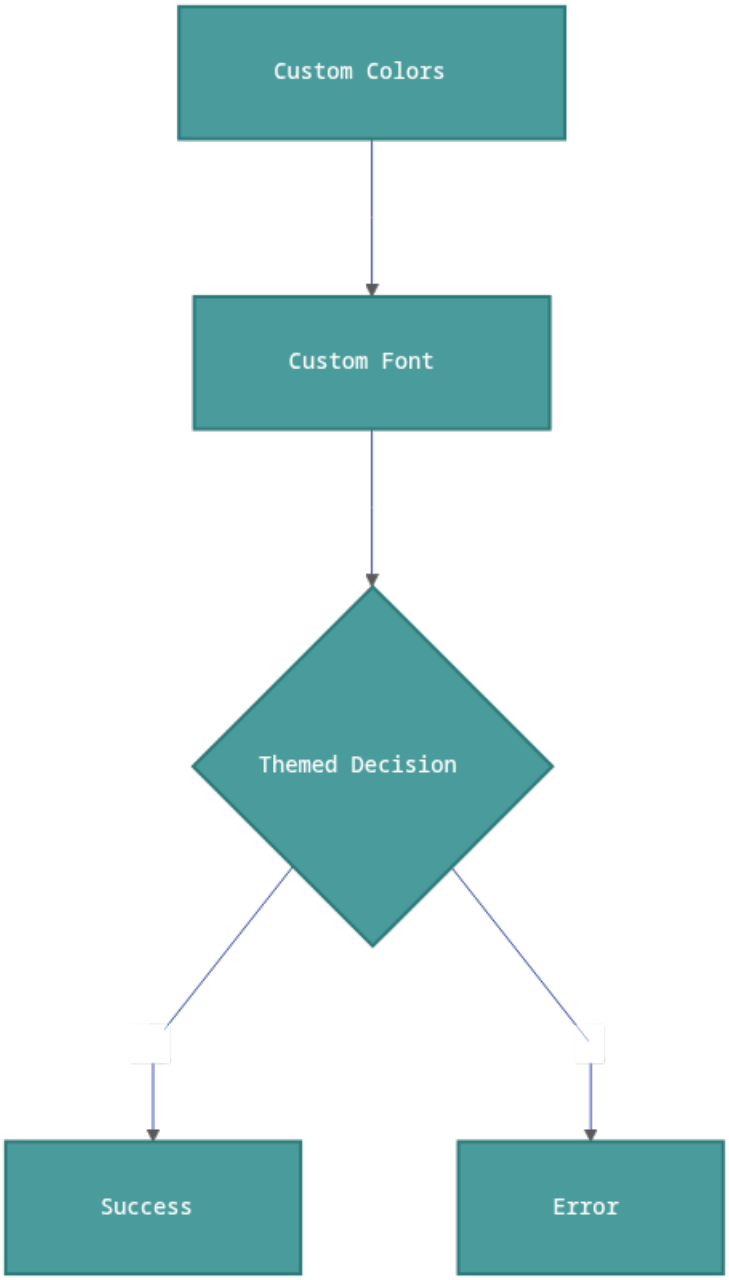
Pre-built Themes



Mermaid Diagram 8

Available themes: `default`, `dark`, `forest`, `neutral`, `base`

Custom Theme Variables



Mermaid Diagram 9

Common Theme Variables:

```
{
  'primaryColor': '#color',         // Main nodes
  'primaryTextColor': '#color',     // Text on primary
  'primaryBorderColor': '#color',   // Border on primary
  'secondaryColor': '#color',       // Secondary nodes
  'tertiaryColor': '#color',        // Tertiary nodes
  'lineColor': '#color',            // Connector lines
  'fontSize': '16px',               // Base font size
  'fontFamily': 'sans-serif',       // Font family
  'background': '#color',           // Diagram background
  'mainBkg': '#color',              // Node background
  'textColor': '#color',            // Default text color
  'edgeLabelBackground': '#color'   // Label background
}
```

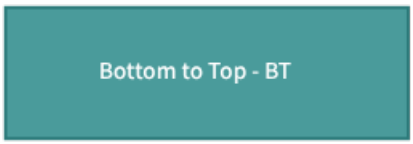
Layout and Direction

Control diagram orientation and flow.

Mermaid Diagram 10

Mermaid Diagram 11

Mermaid Diagram 12



Mermaid Diagram 13

Direction Codes:

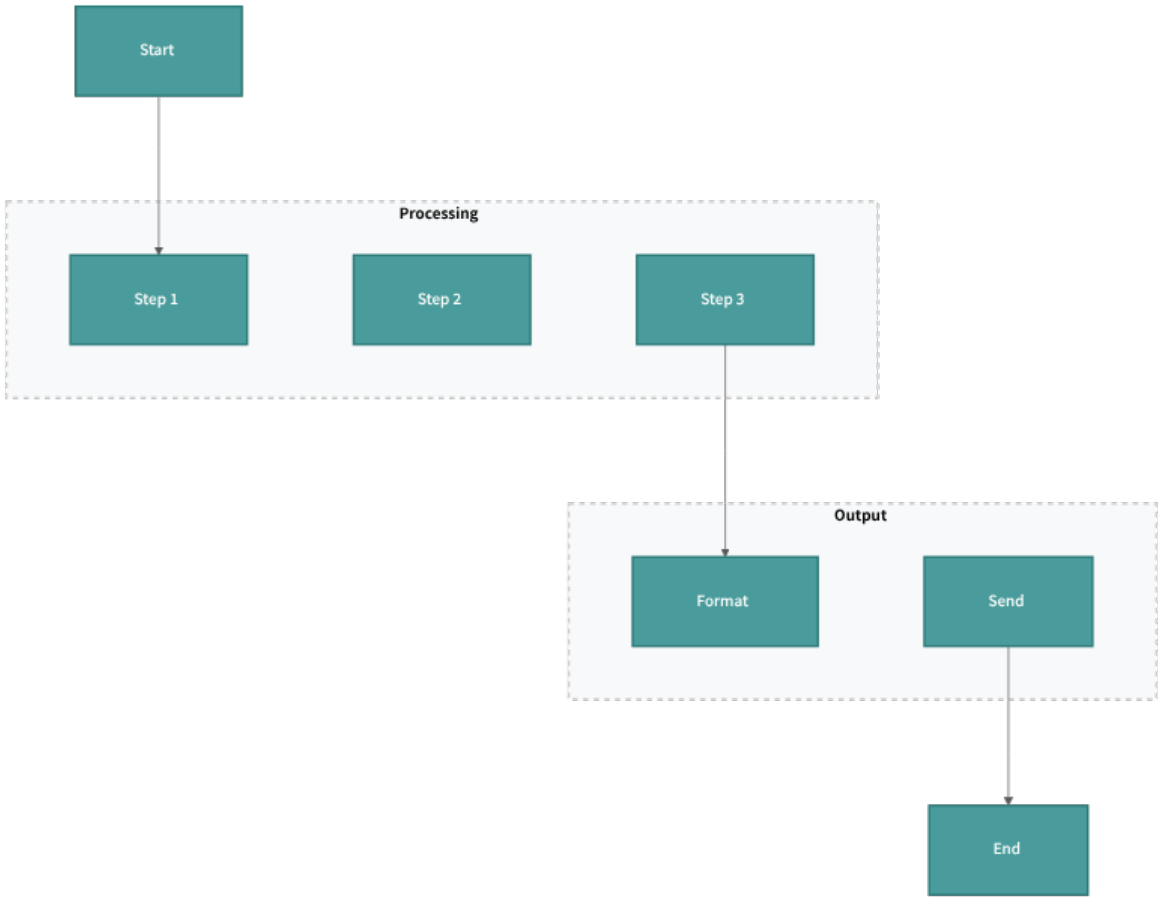
- `TD` or `TB` — Top Down / Top to Bottom (default)
- `LR` — Left to Right
- `RL` — Right to Left

- `BT` — Bottom to Top

Subgraphs for Grouping

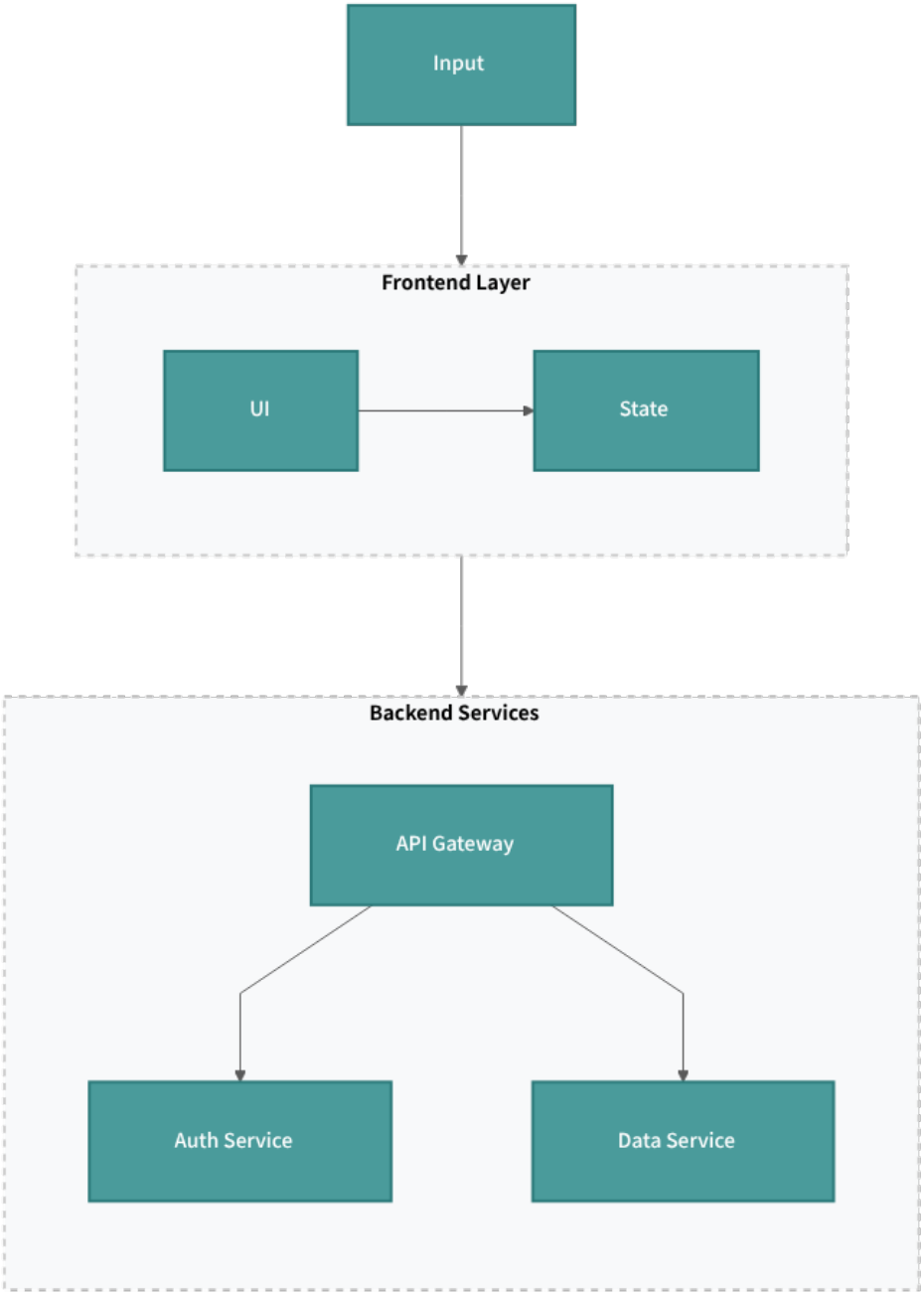
Subgraphs visually group related nodes and can have their own direction.

Basic Subgraph



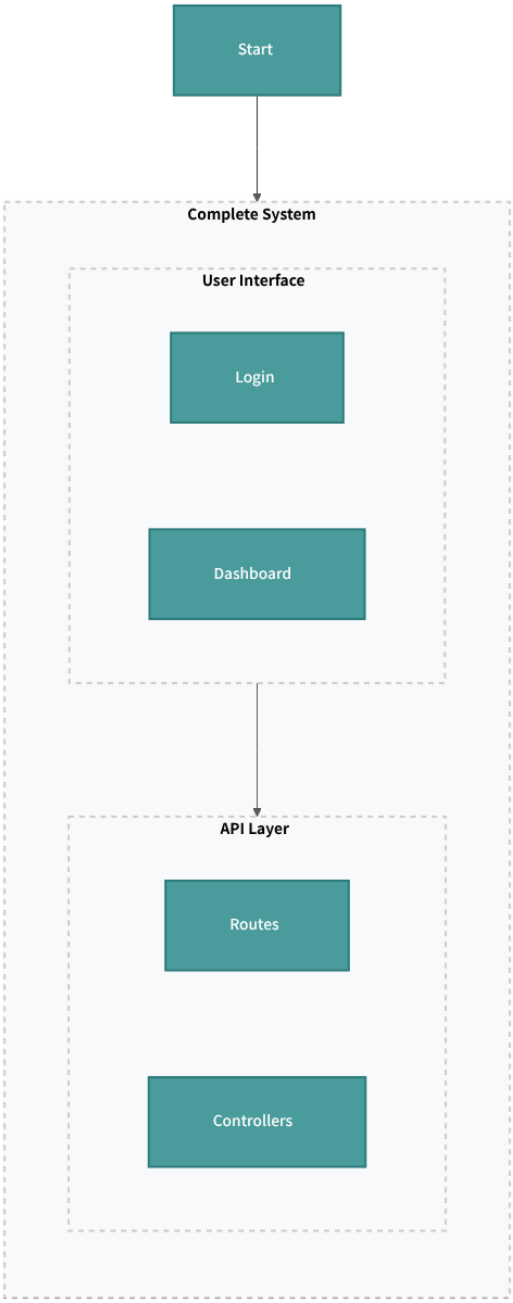
Mermaid Diagram 14

Named Subgraphs with Custom Direction



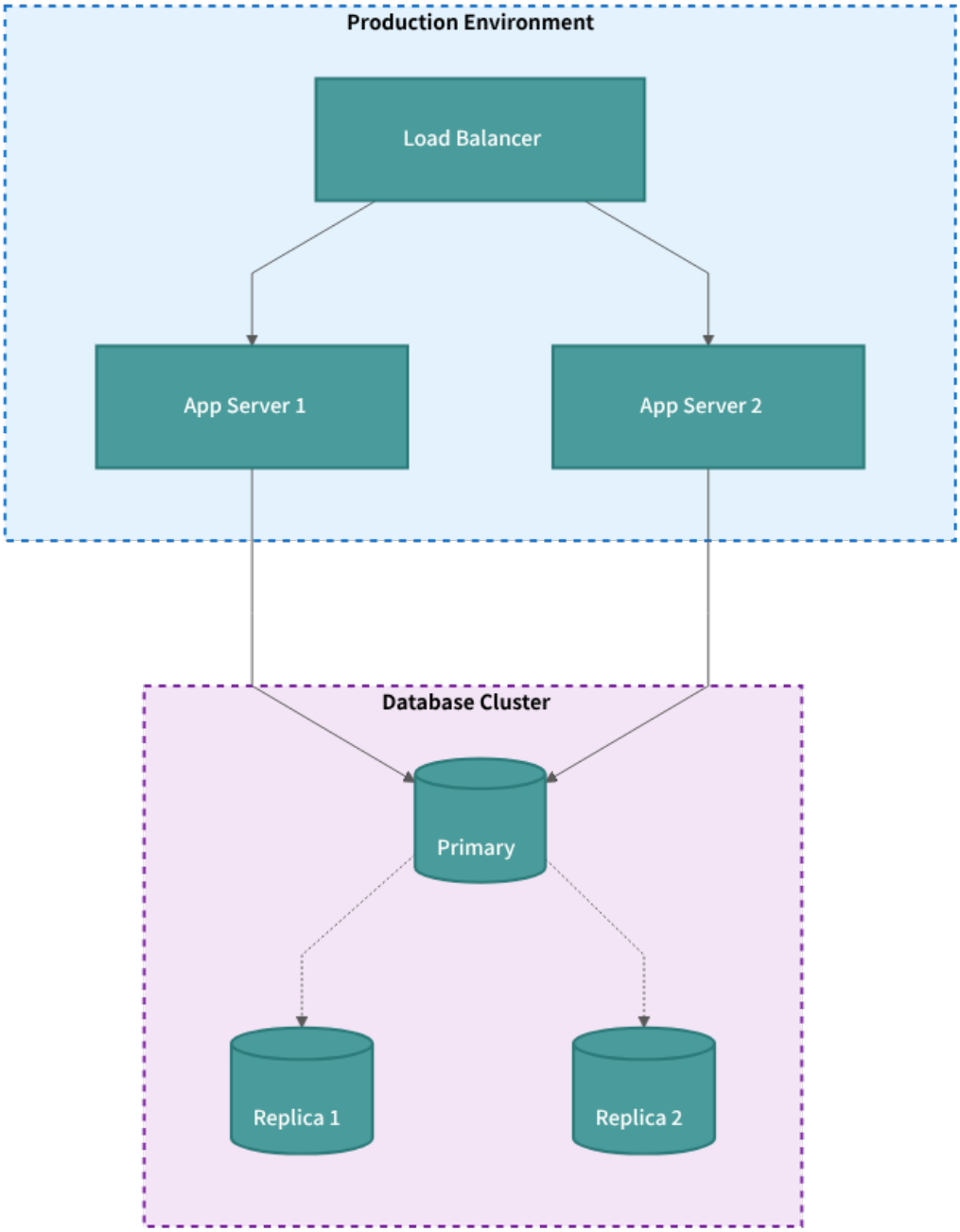
Mermaid Diagram 15

Nested Subgraphs



Mermaid Diagram 16

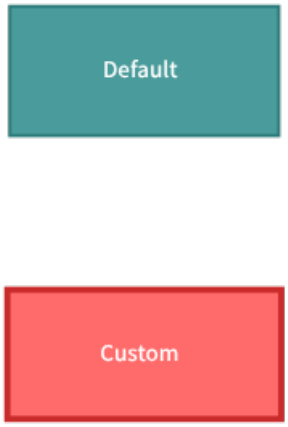
Styled Subgraphs



Mermaid Diagram 17

Advanced Styling Patterns

Individual Node Styling



Mermaid Diagram 18

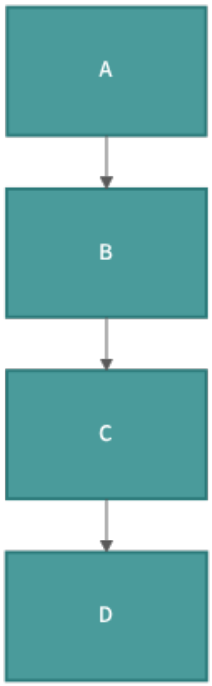
Link Styling by Index

Mermaid Diagram 19

Link Styling Properties:

```
linkStyle <index> stroke:<color>,stroke-width:<width>,stroke-dasharray:<pattern>
```

Default Styles for All Nodes/Links



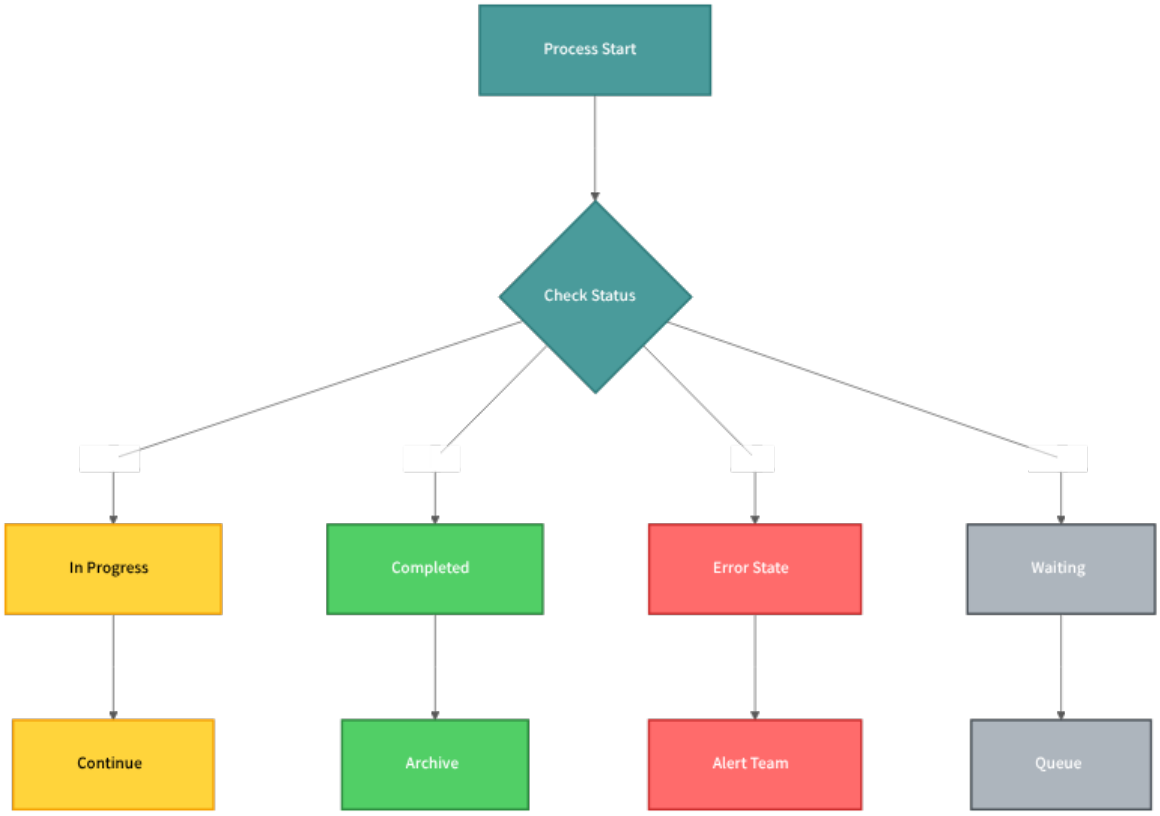
Mermaid Diagram 20

Curve Options:

- `basis` — Smooth curves
- `linear` — Straight lines
- `step` — Step-like connections
- `stepBefore` — Step before
- `stepAfter` — Step after

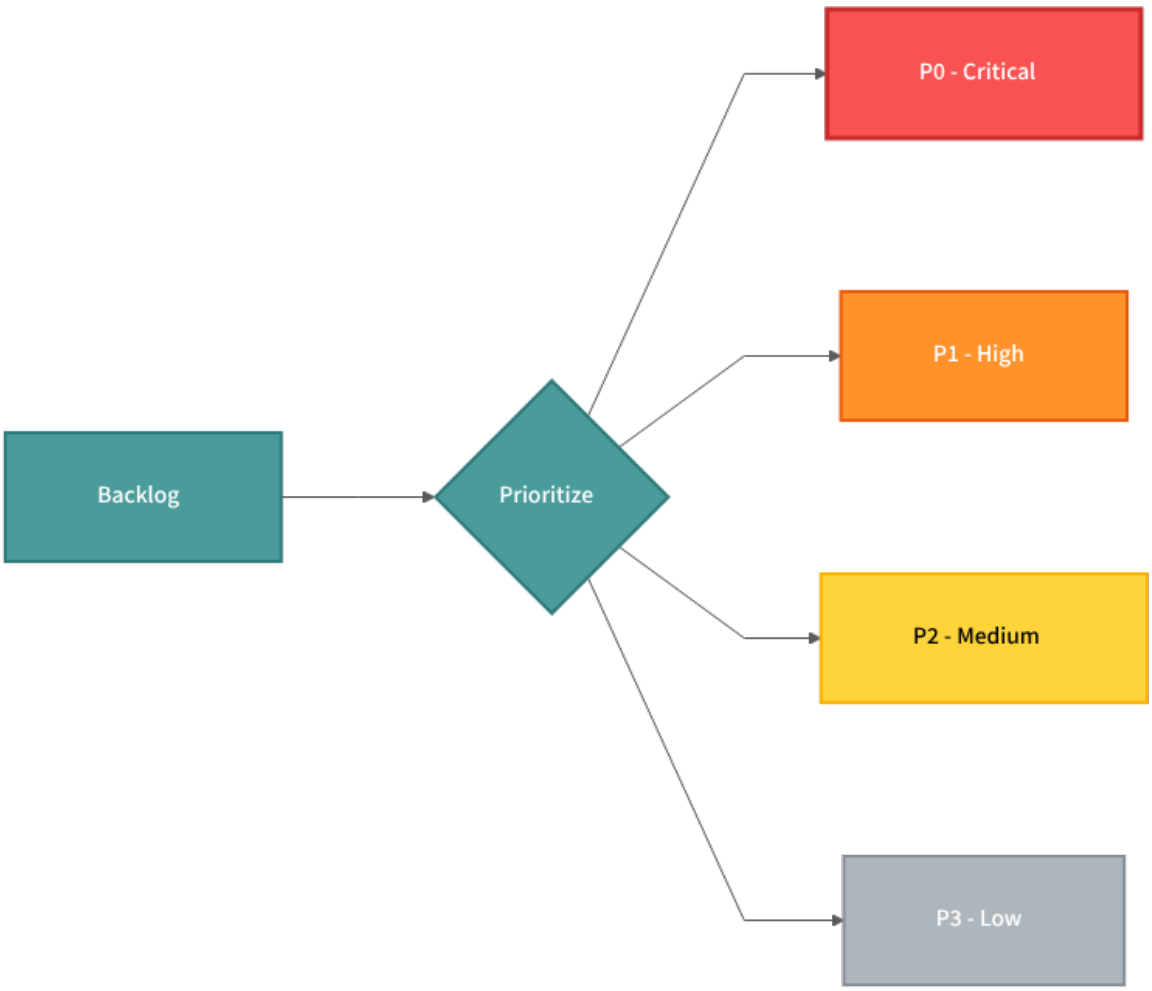
Practical Styling Patterns

Status Indicators



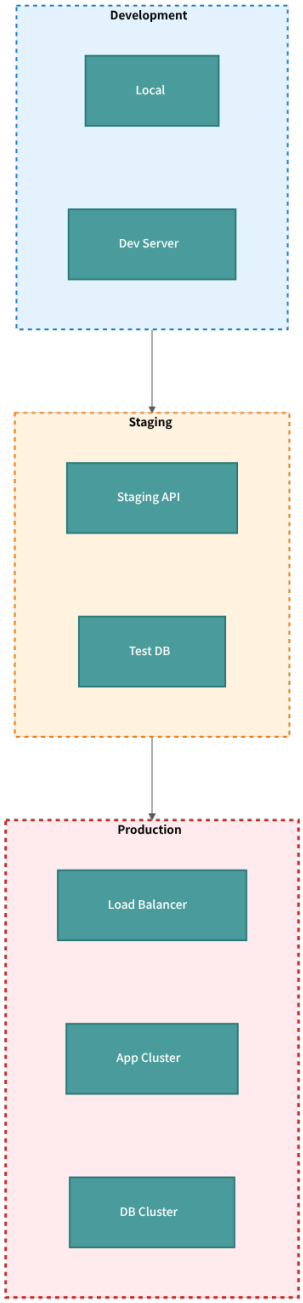
Mermaid Diagram 21

Priority Levels



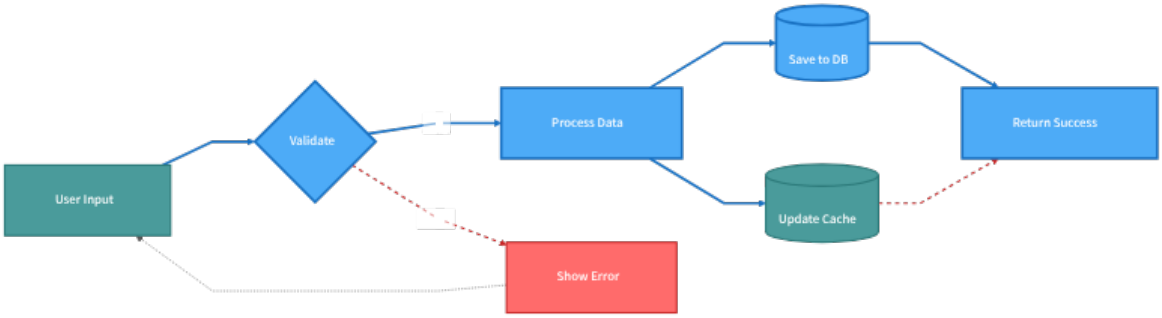
Mermaid Diagram 22

Environment Layers



Mermaid Diagram 23

Data Flow with Emphasis



Mermaid Diagram 24

Interactive Features

Clickable Nodes

> ***Note:*** *Diagram 25 failed to render (unsupported syntax)*

Click Syntax:

```
click <nodeId> "<url>" "<tooltip>" [_blank/_self/_parent/_top]
click <nodeId> href "<url>" "<tooltip>"
click <nodeId> call <function>() "<tooltip>"
```

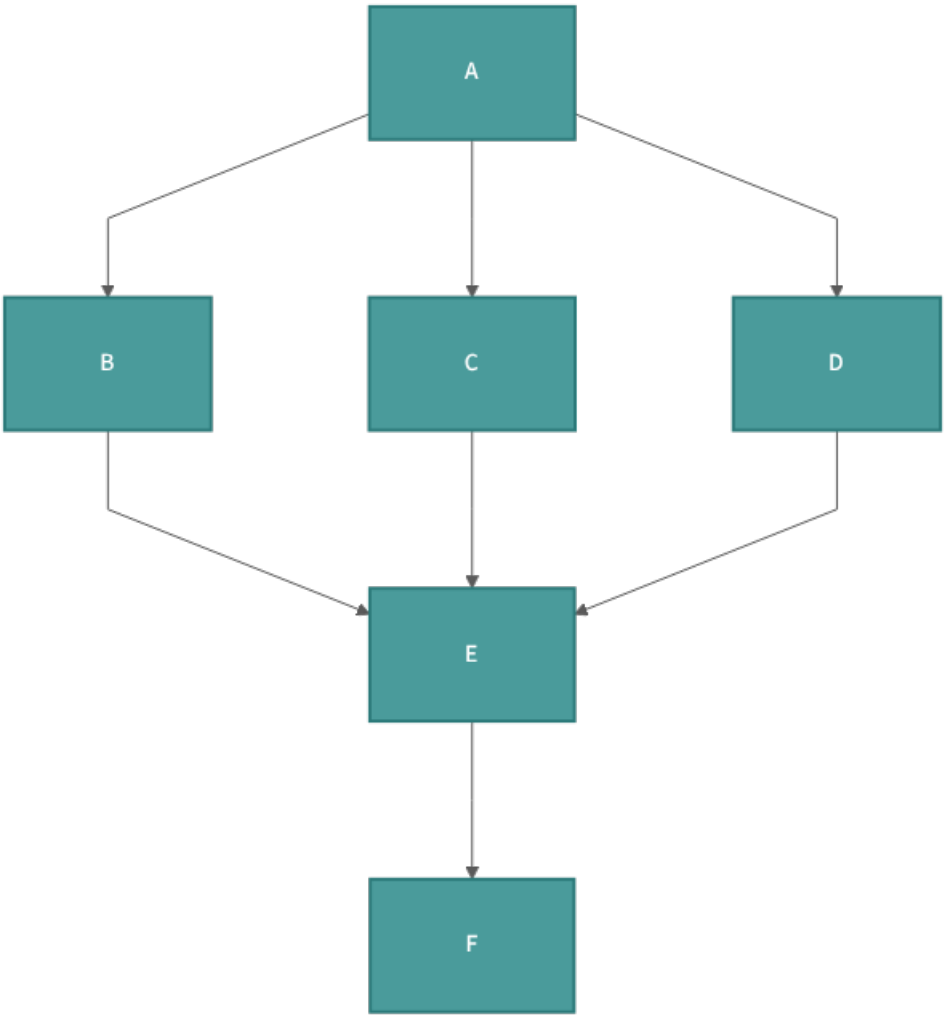
Tooltips (via title attribute)

> ***Note:*** *Diagram 26 failed to render (unsupported syntax)*

Layout Optimization

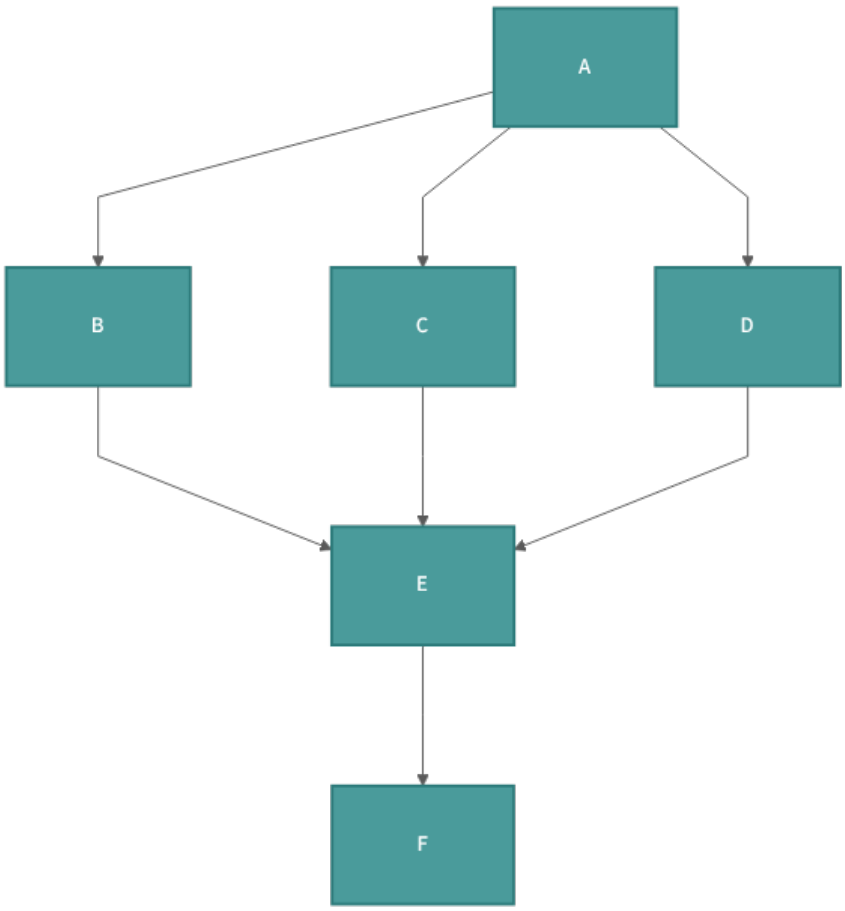
Managing Complex Layouts

Problem: Messy crossing lines



Mermaid Diagram 27

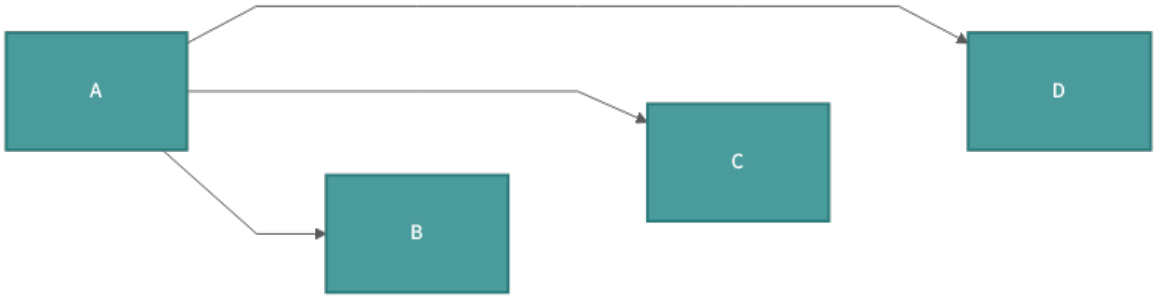
Solution: Use invisible nodes for routing



Mermaid Diagram 28

Rank Separation Control

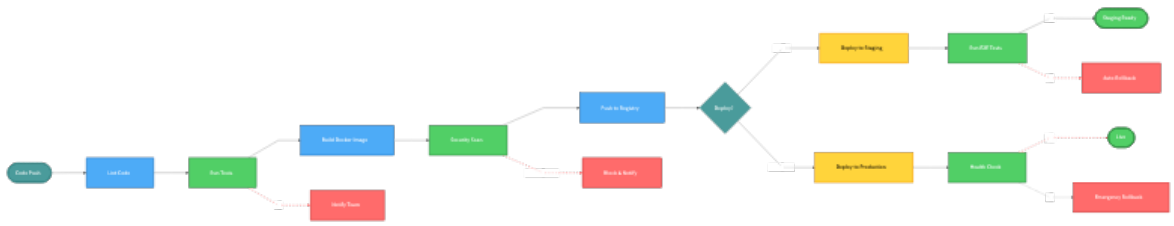
Force nodes onto the same level:



Mermaid Diagram 29

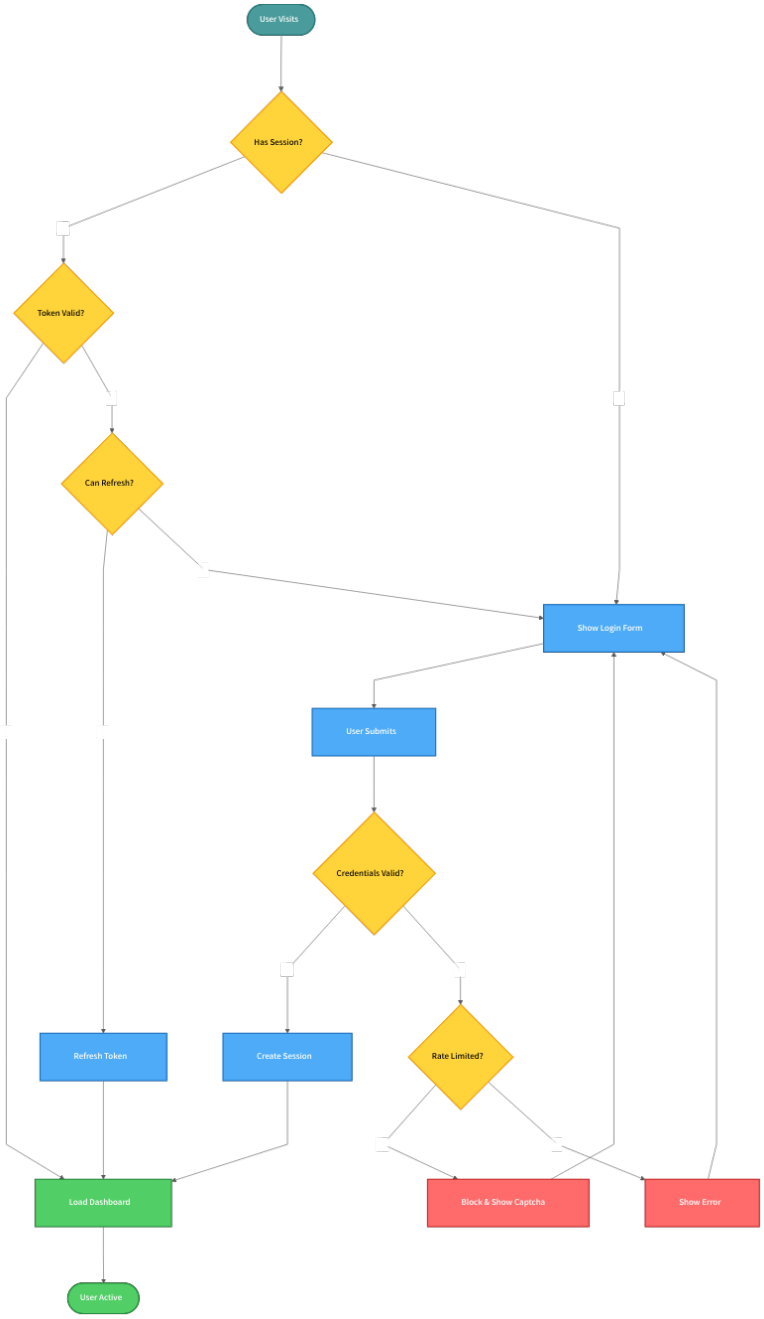
Real-World Examples

CI/CD Pipeline



Mermaid Diagram 30

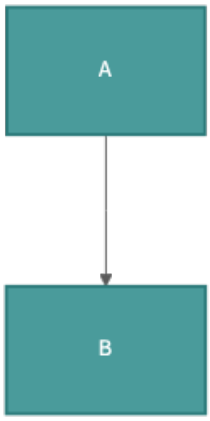
Authentication Flow with Error Handling



Mermaid Diagram 31

Debugging Tips

1. **Test incrementally** — Add nodes/styling one at a time
2. **Use mermaid.live** for real-time preview
3. **Comment out sections** to isolate issues:



Mermaid Diagram 32

4. **Validate class names** — must be alphanumeric
5. **Check link indices** — `linkStyle 0` is the first arrow, counting from top

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Cheat Sheet: Property Values

Colors: - Hex: `#ff6b6b` - RGB: `rgb(255, 107, 107)` - RGBA: `rgba(255, 107, 107, 0.5)` - Named: `red`, `blue`, `green`

Sizes: - Pixels: `2px`, `10px` - Percent: `50%` - Em: `1.5em`

Stroke Patterns: - Solid: `stroke-width:2px` - Dashed: `stroke-dasharray: 5 5` - Dotted: `stroke-dasharray: 1 3` - Long dash: `stroke-dasharray: 10 5`

Font Families: - `monospace` - `sans-serif` - `serif` - `'Inter', system-ui, sans-serif`

Common Color Palettes

Status Colors (Friendly)

```
Success: fill:#51cf66,stroke:#2b8a3e,color:#fff
Warning: fill:#ffd43b,stroke:#f59f00,color:#000
Error: fill:#ff6b6b,stroke:#c92a2a,color:#fff
Info: fill:#4dabf7,stroke:#1971c2,color:#fff
Neutral: fill:#adb5bd,stroke:#495057,color:#fff
```

Priority Levels

```
P0 Critical: fill:#fa5252,stroke:#c92a2a,color:#fff
P1 High: fill:#ff922b,stroke:#e8590c,color:#fff
P2 Medium: fill:#ffd43b,stroke:#fab005,color:#000
P3 Low: fill:#94d82d,stroke:#5c940d,color:#000
```

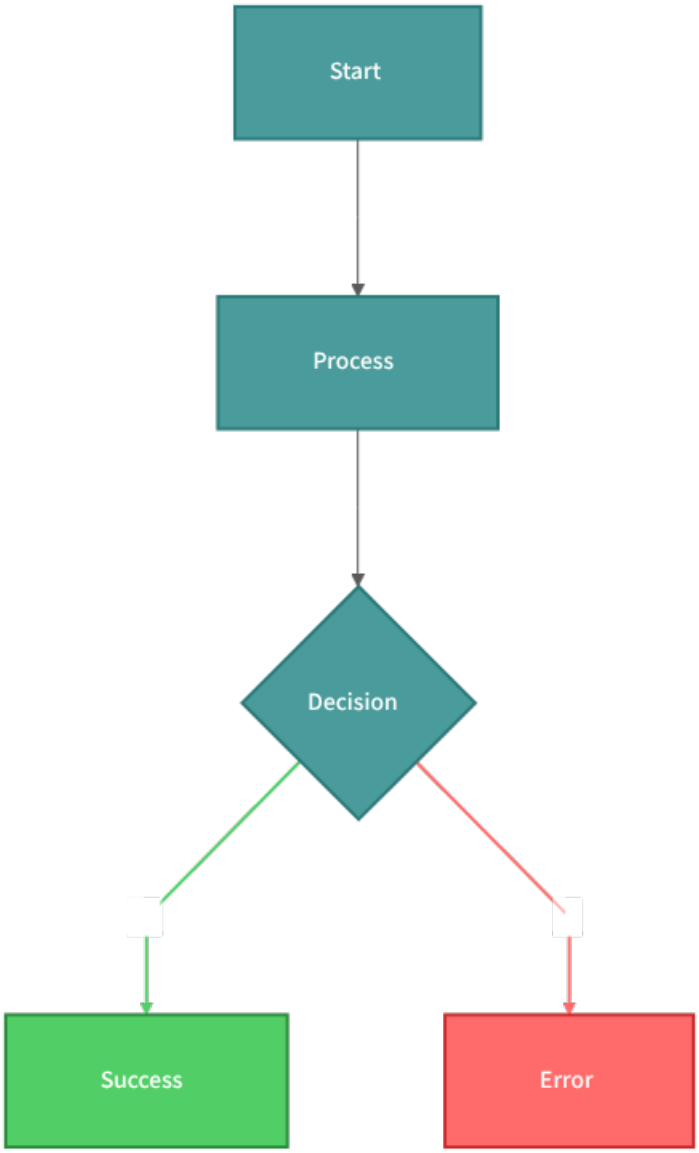
Environment Tiers

```
Dev: fill:#e3f2fd,stroke:#1976d2,color:#000
Staging: fill:#fff3e0,stroke:#f57c00,color:#000
Production: fill:#ffebee,stroke:#c62828,color:#000
```

Quick Customization Workflow

1. **Build structure** (nodes + arrows)
2. **Define classDef** for common styles
3. **Apply classes** to nodes
4. **Style links** if needed
5. **Add subgraphs** for grouping
6. **Apply theme** if needed
7. **Test and refine**

Template:



Mermaid Diagram 33